Human Physiology · PHYS-UG-MEMB-01

PHYS-MEMB-1 Membrane Potentials and the Action Potential

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

PHYS-UG-MEMB-01-C01-BD1

Brain dump 1 · foundational retrieval: Draw the resting membrane and label what sets it

A competent response contains

· A typical neuron rests near minus 70 millivolts, and the minus sign means the inside of the cell is negative compared with the outside.

· Potassium and the large negatively charged proteins are concentrated inside the cell; sodium and chloride are concentrated outside.

· At rest the membrane is far more permeable to potassium than to sodium because potassium leak channels are open, so potassium drifts out down its gradient and leaves negative charge behind.

· Few sodium channels are open at rest, so sodium cannot follow its strong inward gradient in large amounts.

· Negatively charged proteins and phosphates are trapped inside because they are too large to cross the membrane, which adds to the inside negativity.

· Resting membrane potential is the steady voltage difference across the membrane of a cell that is not sending a signal.

Distinctions that matter

· Permeability is not the same as gradient. Sodium has the larger driving force at rest, but potassium has the open channels, so potassium decides the resting voltage.

· Leak channels are always open; voltage gated channels are closed at rest.

Common misconceptions

· Thinking the inside is negative because there is no sodium inside. There is sodium inside, just much less than outside, and the negativity comes from potassium leaving plus the trapped proteins.

· Thinking the resting potential means no ions are moving. Ions move constantly at rest; the movements balance.

Where it is taught: Section 1, resting membrane potential of the PHYS-MEMB-1 Membrane Potentials and the Action Potential notes.

PHYS-UG-MEMB-01-C01-BD2

Brain dump 2 · integration and application: Move the resting potential and say which way

A competent response contains

· Raising outside potassium shrinks the potassium gradient, so less potassium leaves, less negative charge is left behind, and the membrane moves closer to zero, which is depolarization.

· Opening extra potassium leak channels lets more potassium out, so the inside becomes more negative and the membrane moves further below zero, which is hyperpolarization.

· Opening a few sodium channels lets sodium in down its gradient, adding positive charge inside and depolarizing the membrane.

· Removing all outside sodium changes the resting potential very little, because the resting membrane barely lets sodium through in the first place.

Distinctions that matter

· Depolarization means moving toward zero and beyond; hyperpolarization means moving further from zero in the negative direction.

· A change in a gradient matters only if there is an open channel for that ion.

Common misconceptions

· Expecting removal of outside sodium to wipe out the resting potential. It cripples the action potential instead, since the rising phase depends on sodium entry.

Where it is taught: Section 1, what shifts the resting potential of the PHYS-MEMB-1 Membrane Potentials and the Action Potential notes.

PHYS-UG-MEMB-01-C02-BD1

Brain dump 1 · foundational retrieval: Draw the sodium potassium pump at work

A competent response contains

· Each cycle moves three sodium ions out of the cell and two potassium ions in, using one molecule of ATP.

· Because three positive charges leave for every two that enter, the pump moves one net positive charge out per cycle and adds a small amount to the inside negativity.

· Sodium is about 145 millimoles per liter outside and about 15 inside; potassium is about 4 millimoles per liter outside and about 140 inside.

· If a sodium channel opened, sodium would rush in; if a potassium channel opened, potassium would drift out.

· A concentration gradient is a difference in the amount of a substance between two places, here across the membrane.

· The sodium potassium pump builds and maintains both gradients and runs continuously.

Distinctions that matter

· The pump is active transport and needs ATP; leak channels and voltage gated channels are passive and need no ATP.

· The pump contributes only a few millivolts directly; its main job is keeping the gradients that everything else depends on.

Common misconceptions

· Crediting the pump with the whole resting potential. Most of the voltage comes from potassium leaving through leak channels.

· Reversing the direction, saying the pump pushes potassium out. It pulls potassium in.

Where it is taught: Section 2, the sodium potassium pump of the PHYS-MEMB-1 Membrane Potentials and the Action Potential notes.

PHYS-UG-MEMB-01-C02-BD2

Brain dump 2 · integration and application: Lose the pump and follow the consequences

A competent response contains

· With the pump stopped, sodium leaks in and accumulates inside while potassium leaks out and falls inside, so both gradients shrink.

· As the potassium gradient shrinks, the resting potential drifts toward zero, meaning the cell slowly depolarizes.

· The cell can still fire immediately after the pump stops, because the gradients are still nearly full and one action potential moves only a tiny fraction of the ions present.

· An hour later the gradients have run down, so there is no longer enough driving force for sodium to produce a full rising phase and firing fails.

· The gradients act as stored energy: the pump spends ATP to build them, and channels later release that stored energy as current when they open.

Distinctions that matter

· Immediate failure versus gradual failure. Blocking the pump does not stop the next action potential; it stops the hundredth one.

· The pump restores gradients over time but does not repolarize the membrane after a spike; potassium channels do that.

Common misconceptions

· Believing the pump reverses each action potential directly. Repolarization is potassium leaving through voltage gated channels, not pump activity.

Where it is taught: Section 2, gradients as stored energy of the PHYS-MEMB-1 Membrane Potentials and the Action Potential notes.

PHYS-UG-MEMB-01-C02-PR01

Prediction exercise: Predict what rising outside potassium does

A competent response contains

· The potassium concentration gradient decreases, because the outside value rises toward the inside value.

· Potassium leak out of the cell at rest decreases, since the driving force pushing it out is smaller.

· The resting membrane potential moves closer to zero, which is depolarization.

· The membrane sits closer to threshold, so a smaller stimulus can trigger firing at first.

· If depolarization is large or lasting, the number of sodium channels able to open decreases, because sustained depolarization leaves many of them inactivated.

Distinctions that matter

· A mild rise in outside potassium makes cells easier to excite; a large or sustained rise makes them harder to excite and eventually unexcitable.

· Closer to threshold is not the same as more able to fire, because channel availability also matters.

Common misconceptions

· Assuming any depolarization always increases excitability. Past a point the sodium channels sit inactivated and excitability falls.

· Thinking raising outside potassium drives potassium into the cell against everything else. It mainly slows the outward leak.

Where it is taught: Section 2, potassium and membrane excitability of the PHYS-MEMB-1 Membrane Potentials and the Action Potential notes.

PHYS-UG-MEMB-01-C03-BD1

Brain dump 1 · foundational retrieval: Two kinds of electrical signal, side by side

A competent response contains

· A graded potential is a small local voltage change, usually in dendrites or the cell body, produced when chemically gated or mechanically gated channels open.

· Graded potentials vary in size with stimulus strength, are usually a few millivolts, and fade as they spread away from where they started.

· An action potential happens at the axon and travels its full length without fading, swinging from about minus 70 millivolts to about plus 30 millivolts.

· Action potentials are produced by voltage gated sodium and potassium channels and occur only if the membrane reaches threshold.

· Threshold is the voltage, near minus 55 millivolts, at which enough voltage gated sodium channels open that depolarization becomes self sustaining.

· All or nothing means that once threshold is reached the action potential goes to full size every time, and below threshold nothing happens; it applies to the action potential only.

Distinctions that matter

· Graded potentials decide whether a signal starts; the action potential is the signal that gets sent.

· Size varies in graded potentials, while frequency varies in action potentials.

Common misconceptions

· Calling a small depolarization a weak action potential. A subthreshold depolarization is a graded potential, not a small spike.

· Thinking the action potential shrinks as it travels down the axon. It is regenerated along the way and arrives full size.

Where it is taught: Section 3, graded potentials and action potentials of the PHYS-MEMB-1 Membrane Potentials and the Action Potential notes.

PHYS-UG-MEMB-01-C03-BD2

Brain dump 2 · integration and application: From small inputs to a fired axon

A competent response contains

· Three inputs spaced far apart in time each fade before the next arrives, so the membrane never reaches threshold and no action potential fires.

· Three inputs arriving almost together at one spot add up in time, which is temporal summation, and can reach threshold.

· Three inputs arriving together at separate spots add up in space, which is spatial summation, and can also reach threshold.

· A hyperpolarizing input subtracts from the depolarizing ones, so the sum may stay below threshold and no action potential fires.

· A stronger stimulus raises the firing frequency and may recruit more neurons; the height of each action potential does not change, so frequency codes stimulus strength.

Distinctions that matter

· Temporal summation is many inputs close in time at one site; spatial summation is inputs at different sites at the same time.

· Excitatory inputs depolarize, inhibitory inputs hyperpolarize, and the axon start responds to the running total.

Common misconceptions

· Thinking a bigger stimulus produces a bigger action potential. It produces more action potentials per second.

Where it is taught: Section 3, summation and coding of stimulus strength of the PHYS-MEMB-1 Membrane Potentials and the Action Potential notes.

PHYS-UG-MEMB-01-C03-CP01

Comparison table: Graded potential compared with action potential

A competent response contains

· Location: graded potential in dendrites and cell body; action potential at the axon, starting where the axon leaves the cell body.

· Trigger: graded potential from chemically gated, mechanically gated or light gated channels; action potential from voltage gated channels once threshold is reached.

· Ions: graded potentials commonly involve sodium in, potassium out or chloride in; the action potential is sodium in then potassium out.

· Size: graded potentials are a few millivolts; the action potential is about 100 millivolts from rest to peak.

· Stimulus strength: graded potential size varies with it; action potential size does not.

· Travel: graded potentials fade within a short distance; action potentials travel the full axon without fading.

· Threshold: not required for a graded potential, required for an action potential.

· Refractory period: none for graded potentials, present for action potentials.

· Coding: graded potentials code strength by amplitude, action potentials code it by frequency.

Distinctions that matter

· Both signals are changes in membrane voltage, so the difference lies in the channels, the size, the distance and the all or nothing rule.

Common misconceptions

· Filling in refractory period for the graded potential row. Graded potentials can be summed immediately, which is exactly why summation works.

Where it is taught: Section 3, comparison table of the PHYS-MEMB-1 Membrane Potentials and the Action Potential notes.

PHYS-UG-MEMB-01-C04-BD1

Brain dump 1 · foundational retrieval: Draw the action potential and label every gate

A competent response contains

· Key voltages are rest near minus 70 millivolts, threshold near minus 55 millivolts, peak near plus 30 millivolts, and the undershoot near minus 80 millivolts.

· Rising phase: voltage gated sodium channels open and sodium rushes into the cell down both its concentration gradient and the electrical pull.

· Peak: sodium channels inactivate and voltage gated potassium channels are opening.

· Falling phase: potassium leaves the cell through voltage gated potassium channels while sodium entry has stopped.

· Undershoot: the slow potassium channels stay open a little too long, so extra potassium leaves and the membrane dips below the resting potential before the channels close.

· The voltage gated sodium channel has an activation gate and an inactivation gate. At rest the activation gate is closed and the inactivation gate is open; at threshold the activation gate opens and sodium flows; at the peak the inactivation gate swings shut; during the falling phase the activation gate closes again and the inactivation gate reopens only once the membrane has repolarized.

Distinctions that matter

· Closed and inactivated are different states. A closed channel can open; an inactivated one cannot until the membrane repolarizes.

· Sodium channels are fast and potassium channels are slow, which is why the rising phase is steep and the recovery is drawn out.

Common misconceptions

· Saying potassium enters during repolarization. Potassium leaves, which is why the inside becomes negative again.

· Saying the pump causes repolarization. Open potassium channels do.

Where it is taught: Section 4, phases of the action potential of the PHYS-MEMB-1 Membrane Potentials and the Action Potential notes.

PHYS-UG-MEMB-01-C04-BD2

Brain dump 2 · integration and application: Change one gate and redraw the tracing

A competent response contains

· Sodium inactivation gates held open: sodium keeps entering, so the membrane stays depolarized near the peak and the action potential is greatly prolonged instead of returning to rest promptly.

· Slowed potassium channels: the falling phase begins later and takes longer, so the action potential is wider, and the undershoot is delayed.

· Half the sodium channels blocked: the rising phase is slower and the peak is lower, and a weak stimulus may no longer reach threshold at all.

· Outside sodium halved: the inward driving force for sodium falls, so the peak is lower, while threshold may still be reached if enough channels open.

Distinctions that matter

· Changes in sodium affect the rising phase and the peak; changes in potassium affect the falling phase and the undershoot.

· A lower peak is a problem of driving force or channel number; a longer spike is a problem of gate timing.

Common misconceptions

· Expecting a partial sodium block to shrink every action potential a little. Spikes that still reach threshold are close to full size; the stronger effect is that fewer stimuli succeed at all.

Where it is taught: Section 4, channel changes and the tracing of the PHYS-MEMB-1 Membrane Potentials and the Action Potential notes.

PHYS-UG-MEMB-01-C04-RT01

Retrieval exercise: Rebuild the action potential step by step

A competent response contains

· 1. A stimulus opens gated channels and depolarizes the membrane toward threshold.

· 2. At threshold near minus 55 millivolts, voltage gated sodium activation gates open.

· 3. Sodium moves into the cell, and that depolarization opens still more sodium channels, which is a positive feedback cycle.

· 4. The membrane rises steeply to about plus 30 millivolts.

· 5. Sodium inactivation gates close, ending sodium entry.

· 6. Slow voltage gated potassium channels finish opening.

· 7. Potassium moves out of the cell and the membrane repolarizes toward the resting potential.

· 8. Potassium channels close late, so extra potassium leaves and the membrane undershoots to about minus 80 millivolts.

· 9. Potassium channels close, sodium inactivation gates reopen, and leak channels plus the sodium potassium pump return the membrane to minus 70 millivolts.

Distinctions that matter

· The positive feedback of step 3 is what makes the action potential all or nothing.

· Inactivation of sodium channels and opening of potassium channels are separate events that happen at about the same time.

Common misconceptions

· Putting potassium channel opening before sodium inactivation as the cause of the peak. Both contribute, but the peak is set by sodium entry stopping.

Where it is taught: Section 4, sequence of channel events of the PHYS-MEMB-1 Membrane Potentials and the Action Potential notes.

PHYS-UG-MEMB-01-C04-CA01

Clinical application: A numb tooth after a dental injection

A competent response contains

· Recognize: the drug blocks the action potential itself, not the resting potential and not the graded potential, so the sensory neuron can still respond locally but cannot send a signal.

· Explain: a blocked voltage gated sodium channel cannot pass sodium, so the rising phase never develops; without a rising phase the membrane cannot depolarize the next patch of axon to threshold, so nothing is regenerated along the axon and no signal reaches the brain.

· Differentiate: small diameter axons, including the thinly myelinated and unmyelinated pain fibers, are blocked first because they have less membrane to block and a smaller safety margin, while the large myelinated axons that carry pressure and the motor axons that move the jaw keep conducting longer.

· Investigate: touch the gum with a sharp probe and ask whether it feels sharp or only like pressure, and compare the treated side with the untreated side using the same light touch; loss of sharpness with pressure preserved shows the block has reached the pain fibers but not the large touch fibers.

· Decide: yes, the resting membrane potential of the blocked neuron is still normal.

· Justify: the resting potential is set by potassium leak channels, the trapped negative proteins and the sodium potassium pump, and none of those are affected by a drug that binds only voltage gated sodium channels, so the cell still rests near minus 70 millivolts while being unable to fire.

Distinctions that matter

· A cell can have a perfectly normal resting potential and still be unable to produce an action potential.

· Fiber size and myelination explain the order in which sensations disappear.

Common misconceptions

· Thinking the anesthetic kills the neuron. The block is reversible, which is why normal sensation returns in about two hours.

· Thinking the drug blocks the brain's perception rather than the axon's signal.

Where it is taught: Section 4, voltage gated sodium channels, with Section 6 for fiber size of the PHYS-MEMB-1 Membrane Potentials and the Action Potential notes.

PHYS-UG-MEMB-01-C05-BD1

Brain dump 1 · foundational retrieval: The two refractory periods on one tracing

A competent response contains

· The absolute refractory period covers the rising phase and most of the falling phase; no stimulus of any strength can produce a second action potential during it.

· During the absolute refractory period the sodium activation gates have opened and the inactivation gates are closed, and the closed inactivation gate is the reason no second spike is possible.

· The relative refractory period covers the end of the falling phase and the undershoot; a second action potential is possible but needs a stronger than usual stimulus.

· During the relative refractory period potassium channels are still open, so potassium is still leaving and the membrane is more negative than rest, which puts threshold further away.

· Sodium channels must reset, meaning the inactivation gate reopens while the activation gate closes, and that reset requires the membrane to repolarize back toward the resting range near minus 70 millivolts.

Distinctions that matter

· Absolute means impossible; relative means harder.

· The absolute period is about channel state, while the relative period is mostly about voltage being further from threshold.

Common misconceptions

· Blaming the refractory period on depleted ion gradients. One action potential moves very few ions; the cause is channel state and membrane voltage.

Where it is taught: Section 5, refractory periods of the PHYS-MEMB-1 Membrane Potentials and the Action Potential notes.

PHYS-UG-MEMB-01-C05-BD2

Brain dump 2 · integration and application: Why the signal runs one way and has a speed limit

A competent response contains

· The patch just behind the advancing spike has sodium channels in the inactivated state, so it cannot fire again immediately, and the signal therefore moves only forward.

· The absolute refractory period sets the maximum firing rate, since a neuron cannot fire again until it has passed.

· With an absolute refractory period near one millisecond, the ceiling is roughly 1000 action potentials per second, and real neurons fire well below that.

· Without refractory periods, signals would spread in both directions along axons and summed stimuli would produce continuous uncontrolled firing, so timing information and orderly muscle contraction would be lost.

Distinctions that matter

· One way conduction comes from refractoriness behind the spike, not from any one way property of the channels themselves.

· Maximum firing rate and conduction speed are different things; refractory periods set the first, myelin and diameter set the second.

Common misconceptions

· Thinking the axon is built to carry signals in only one direction. An axon stimulated in the middle in the laboratory conducts both ways; in the body the signal starts at one end.

Where it is taught: Section 5, one way conduction and firing limits of the PHYS-MEMB-1 Membrane Potentials and the Action Potential notes.

PHYS-UG-MEMB-01-C06-BD1

Brain dump 1 · foundational retrieval: Draw a myelinated axon and show how the signal jumps

A competent response contains

· Myelin is a fatty wrapping made by Schwann cells in the peripheral nervous system and by oligodendrocytes in the central nervous system.

· Myelin insulates the membrane so charge does not leak out, and it leaves the covered membrane with very few open channels.

· Nodes of Ranvier are the bare gaps between myelin segments, and voltage gated sodium channels are clustered there.

· The action potential is regenerated only at the nodes and the depolarization spreads quickly through the cytoplasm under each myelin segment, so the signal appears to jump from node to node; this is saltatory conduction.

· Conduction is faster in axons that are myelinated and in axons that are wider.

· A large myelinated axon conducts at roughly 100 meters per second, while a small unmyelinated axon conducts at roughly 1 meter per second.

Distinctions that matter

· Myelin does not make the action potential bigger; it makes the spread between nodes faster.

· Diameter helps because a wider axon offers less internal resistance to the spread of charge.

Common misconceptions

· Picturing the action potential physically leaping through the air between nodes. The local current flows through the cytoplasm and the spike is regenerated at each node.

· Thinking myelin covers the whole axon continuously. The nodes are essential gaps, not defects.

Where it is taught: Section 6, myelin and saltatory conduction of the PHYS-MEMB-1 Membrane Potentials and the Action Potential notes.

PHYS-UG-MEMB-01-C06-BD2

Brain dump 2 · integration and application: Strip the myelin and compare the cost

A competent response contains

· Losing patches of myelin slows conduction, and where the loss is wide the signal may fail to reach the next node at all, so sensation or movement served by that axon becomes slow, patchy or absent.

· The signal can stop completely because the membrane under the former sheath has very few voltage gated sodium channels, so a bare stretch cannot regenerate the spike and the depolarization leaks away before reaching the next node.

· Myelin gives speed in a thin axon, while widening the axon gives speed at a large cost in space, so myelin lets a nerve pack many fast axons into a small cross section.

· Sharp, well localized pain travels on thinly myelinated axons, and dull aching pain travels on unmyelinated axons, which is why sharp pain arrives first.

Distinctions that matter

· Slowed conduction and failed conduction are different outcomes, set by how much sheath is lost at one place.

· Demyelination leaves the axon itself alive, which is why some function can return if the sheath is repaired.

Common misconceptions

· Assuming loss of myelin cuts the axon. The axon is intact; the insulation is what has gone.

Where it is taught: Section 6, conduction speed and loss of myelin of the PHYS-MEMB-1 Membrane Potentials and the Action Potential notes.

PHYS-UG-MEMB-01-C05-Q01

Board-style question

Correct answer: C. The inactivation gates of the voltage gated sodium channels are closed and will not reopen until the membrane repolarizes

Why it is right: Once the inactivation gate closes at the peak, the channel is unavailable no matter how strong the stimulus, and it resets only after the membrane repolarizes, which is exactly what makes the period absolute.

A is wrong: A single action potential moves only a tiny fraction of the sodium and potassium present, so the gradients are still essentially intact and the pump is not the limiting factor.

B is wrong: A membrane more negative than rest describes the undershoot, and that is the relative refractory period, when a strong stimulus can still work.

D is wrong: The voltage gated potassium channels are open rather than closed during this part of the tracing, and their opening is what repolarizes the membrane.

A competent response contains

· The absolute refractory period exists because the inactivation gates of the voltage gated sodium channels are shut and cannot reopen until the membrane repolarizes.

Distinctions that matter

· Channel state explains the absolute period; a more negative membrane voltage explains the relative period.

Common misconceptions

· Attributing refractoriness to used up gradients or to pump activity.

Where it is taught: Section 5, refractory periods of the PHYS-MEMB-1 Membrane Potentials and the Action Potential notes.

PHYS-UG-MEMB-01-C06-Q02

Board-style question

Correct answer: B. The sheath keeps charge from leaking across the membrane, so the depolarization spreads quickly to the next node of Ranvier, where the sodium channels are clustered

Why it is right: Insulation keeps the local current inside the axon so it spreads rapidly to the next node, where clustered voltage gated sodium channels fire a fresh action potential, and this node to node pattern is saltatory conduction.

A is wrong: Myelin is insulation made of membrane lipid and does not supply ATP; the pump gets its ATP from the axon's own metabolism.

C is wrong: Sodium channels in a myelinated axon are clustered at the nodes rather than spread evenly, and spreading them out evenly would slow conduction.

D is wrong: The absolute refractory period depends on sodium channel gating and is not shortened by myelin, and firing rate is a separate matter from conduction speed.

A competent response contains

· Myelin speeds conduction by preventing charge leak across the covered membrane, so depolarization reaches the next node quickly, and the sodium channels clustered at the nodes regenerate the spike there.

Distinctions that matter

· Speed comes from faster passive spread between nodes, not from a bigger or faster individual action potential.

Common misconceptions

· Thinking myelin adds channels or energy along the axon rather than insulating it.

Where it is taught: Section 6, myelin and saltatory conduction of the PHYS-MEMB-1 Membrane Potentials and the Action Potential notes.